

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

# Chapter 23: Graphical Representation (Histograms and Ogives)

Previous year board questions, topic-wise: 13 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

## 1 Mode from a histogram

1. Draw a histogram for the marks of 164 students given below, and use it to find the mode.

ICSE 2025, 3 marks

| Marks              | 30–40 | 40–50 | 50–60 | 60–70 | 70–80 |
|--------------------|-------|-------|-------|-------|-------|
| Number of students | 10    | 26    | 40    | 54    | 34    |

2. A histogram shows the heights of the plants in a school garden. Its rectangles stand on the classes 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and 90–100 (in cm) and have heights 4, 2, 8, 12, 6, 3 and 4. (a) Make the frequency table. (b) State the modal class. (c) Find the mode. (d) How many plants are between 80 cm and 90 cm tall? ICSE 2024, 5 marks

3. The daily wages of 60 workers are given below. Draw a histogram and use it to find the mode. ICSE 2023, 3 marks

| Daily wage (₹)    | 200–300 | 300–400 | 400–500 | 500–600 | 600–700 | 700–800 |
|-------------------|---------|---------|---------|---------|---------|---------|
| Number of workers | 6       | 10      | 14      | 16      | 10      | 4       |

4. Which of these cannot be found graphically for a grouped frequency distribution? (a) median (b) mode (c) quartiles (d) mean ICSE 2023, 1 mark

5. Draw a histogram for the weekly wages below and estimate the mode from it.

ICSE 2020, 4 marks

| Weekly wages (₹) | 3000–4000 | 4000–5000 | 5000–6000 | 6000–7000 | 7000–8000 | 8000–9000 | 9000–10000 |
|------------------|-----------|-----------|-----------|-----------|-----------|-----------|------------|
| Number of people | 4         | 9         | 18        | 6         | 7         | 2         | 4          |

6. The runs scored by 50 batsmen are given below. Draw a histogram and estimate the mode.

ICSE 2018, 4 marks

| Runs scored       | 3000–4000 | 4000–5000 | 5000–6000 | 6000–7000 | 7000–8000 | 8000–9000 | 9000–10000 |
|-------------------|-----------|-----------|-----------|-----------|-----------|-----------|------------|
| Number of batsmen | 4         | 18        | 9         | 6         | 7         | 2         | 4          |

## 2 Reading an ogive

7. An ogive for the monthly salaries of the workers in a factory passes through the points below. Find (a) the total number of workers (b) the median class (c) the lower-quartile class (d) the number of workers earning at least ₹6,000 but less than ₹10,000. ICSE 2025, 4 marks

| Salary less than (₹)         | 2000 | 4000 | 6000 | 8000 | 10000 | 12000 |
|------------------------------|------|------|------|------|-------|-------|
| Cumulative number of workers | 10   | 20   | 35   | 55   | 70    | 75    |

8. The ages of 100 policy holders are given below. Draw an ogive and use it to find (a) the median age (b) the number of policy holders older than 52 years. ICSE 2024, 4 marks

| Age (years)          | 20–25 | 25–30 | 30–35 | 35–40 | 40–45 | 45–50 | 50–55 | 55–60 |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Policy holders       | 2     | 4     | 12    | 20    | 28    | 22    | 8     | 4     |
| Cumulative frequency | 2     | 6     | 18    | 38    | 66    | 88    | 96    | 100   |

9. The weights of 60 students are given below. Draw an ogive and use it to find (a) the median (b) the upper quartile (c) the number of students who weigh more than 37 kg.

ICSE 2023, 6 marks

| Weight (kg)        | 28–30 | 30–32 | 32–34 | 34–36 | 36–38 | 38–40 | 40–42 | 42–44 |
|--------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Number of students | 2     | 4     | 10    | 13    | 15    | 9     | 5     | 2     |

10. The distances thrown by 40 students in a shot-put competition are given below. Draw an ogive and use it to find (a) the median (b) the upper quartile (c) the number of students who threw more than 16.5 m. ICSE 2020, 6 marks

| Distance (m)       | 12–13 | 13–14 | 14–15 | 15–16 | 16–17 | 17–18 | 18–19 |
|--------------------|-------|-------|-------|-------|-------|-------|-------|
| Number of students | 3     | 9     | 12    | 9     | 4     | 2     | 1     |

- 11.** The marks of 120 students in a test are given below. Draw an ogive and use it to estimate (a) the median (b) the number of students who failed if the pass mark is 50 (c) the upper quartile. **ICSE 2019, 6 marks**

| Marks    | 0–10 | 10–20 | 20–30 | 30–40 | 40–50 | 50–60 | 60–70 | 70–80 | 80–90 | 90–100 |
|----------|------|-------|-------|-------|-------|-------|-------|-------|-------|--------|
| Students | 5    | 9     | 16    | 22    | 26    | 18    | 11    | 6     | 4     | 3      |

- 12.** The heights of 60 boys are given below. Draw an ogive and use it to estimate (a) the median (b) the lower quartile (c) the number of boys taller than 158 cm. **ICSE 2018, 6 marks**

| Height (cm)    | 135–140 | 140–145 | 145–150 | 150–155 | 155–160 | 160–165 | 165–170 |
|----------------|---------|---------|---------|---------|---------|---------|---------|
| Number of boys | 4       | 8       | 20      | 14      | 7       | 6       | 1       |

- 13.** The daily wages of 80 workers are given below. Draw an ogive and use it to estimate (a) the median wage (b) the lower quartile (c) the number of workers who earn more than ₹625 a day. **ICSE 2017, 6 marks**

| Wages (₹)         | 400–450 | 450–500 | 500–550 | 550–600 | 600–650 | 650–700 | 700–750 |
|-------------------|---------|---------|---------|---------|---------|---------|---------|
| Number of workers | 2       | 6       | 12      | 18      | 24      | 13      | 5       |

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